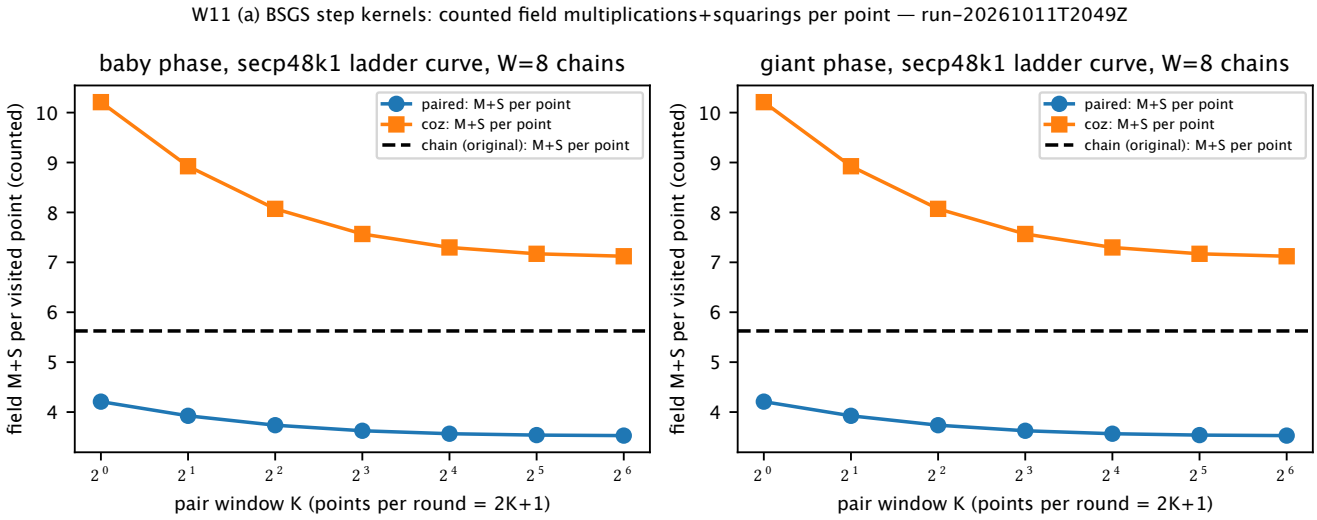
1. Run report: lit2026 W11, run-20261011T2049Z

What ran. Measurements for workstream W11 of the lit2026 speedup program (branch `lit2026/w11-generic-engineering`, stacked on aburan28/crypto#1681). Observations only; interpretation is in the study README / `results.pdf`.

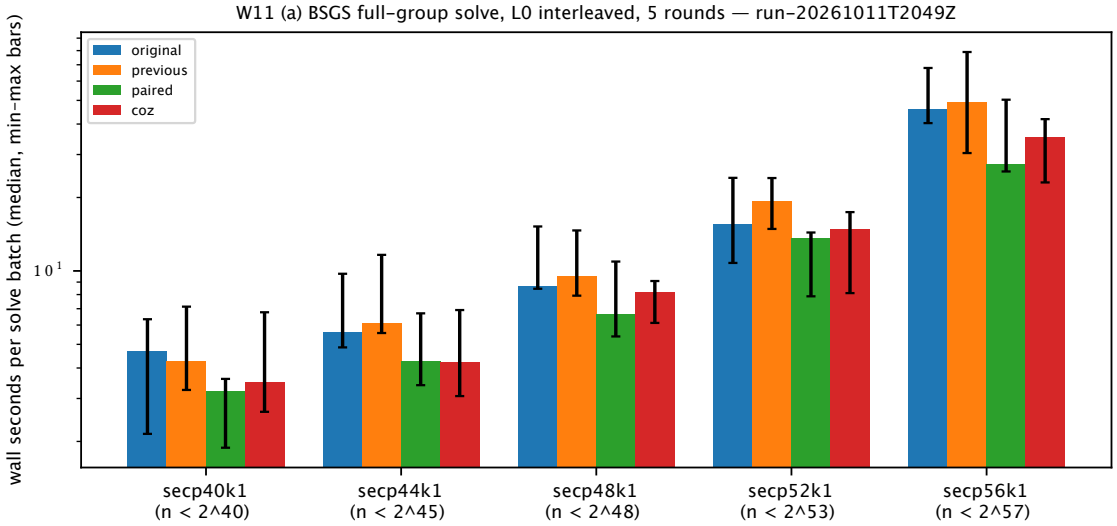
- **Host.** Apple M4 Pro, 14 cores, macOS 26.6, shared with other sessions; 1-minute load 72–155 (recorded with every BSGS and class-walk sample). **Isolation L0.**
- **Arms.** `original` = program baseline `4b6e09b78` (baseline worktree, read-only); `previous/candidate` = one binary of this branch with the switch off/on.
- **Raw records.** `raw/*.jsonl` (one JSON object per sample), `raw/classwalk_19_1/` (samples.tsv, round-1 stdout and snapshots of each arm), `raw/summary.json` (medians, min–max, per-round ratios, identity checks; computed by `examples/lit2026_w11_summarize.rs`). Preliminary runs: `raw/prelim/`.
- **Requested vs completed.** All requested runs completed. One preliminary biext run stopped at `r_bits = 56` (no prime `p = c·r − 1 < 2^62` with `c ≤ 64`); the final run covers 16–52 bits.

1.1. (a) BSGS step kernels

Counted field operations per visited point (`raw/bsgs_opcount.jsonl`, W = 8, giant phase shown in the graph's right panel); every kernel visited the same `(index, x)` pairs as the chain (`same_points_as_chain = true` in every row).



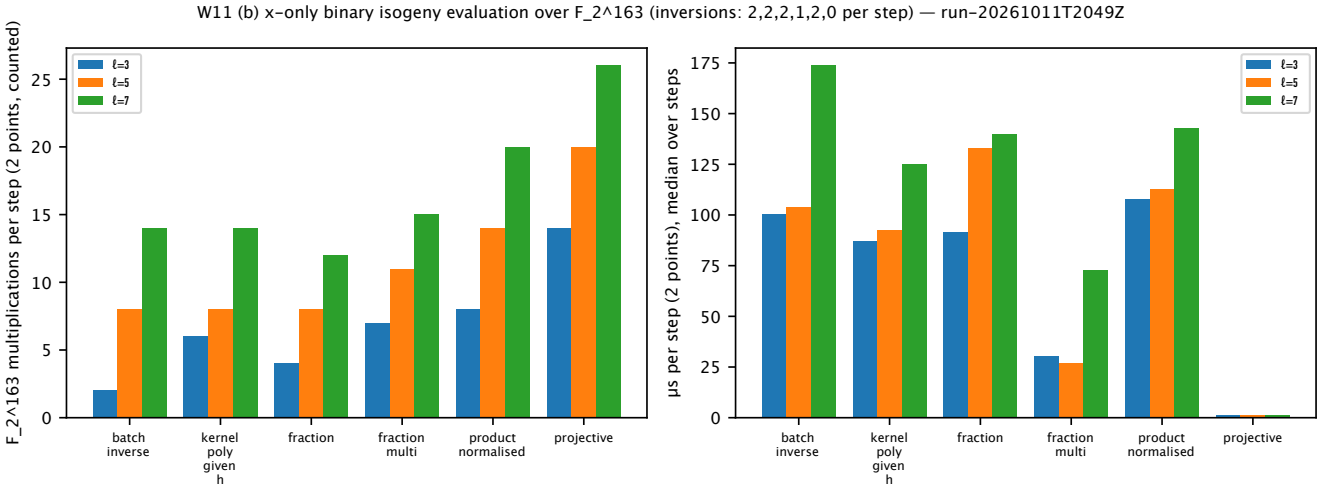
Wall seconds per solve batch, 5 rounds × 4 arms × 5 curves (`raw/bsgs_time.jsonl`); answers digest and table entries identical across arms at every size.



curve	original s	previous s	paired s	coz s	original/paired (min–max rounds)
secp40k1	4.67	4.28	3.20	3.49	1.459 (1.054–1.917)
secp44k1	5.63	6.12	4.27	4.22	1.316 (1.015–2.002)
secp48k1	8.63	9.55	6.64	8.16	1.301 (1.299–1.603)
secp52k1	15.55	19.40	13.58	14.77	1.145 (0.996–1.976)
secp56k1	45.84	48.99	27.44	35.32	1.670 (1.349–1.670)

1.2. (b) x-only binary isogeny evaluation

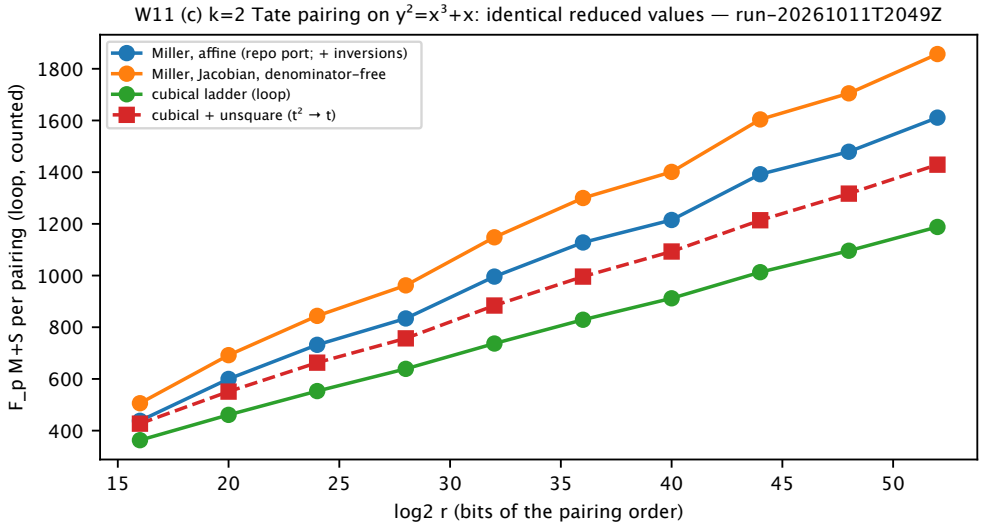
`raw/binary_sect163.jsonl` (ℓ = 3, 5, 7; 8 steps each; 2000 reps × 7 rounds per step), `raw/binary_torsion19.jsonl` (458 kernels, 20 reps × 9 rounds), `raw/classwalk_19_1/` (12 rounds × 3 arms; snapshots identical without `seconds`).



Torsion-walk step: `F_(q^8)` mults 627 460 ⟶ 419 070, inversions 916 ⟶ 458, µs per kernel 1564 ⟶ 1211 (medians; per-round ratios 0.94–1.93). Class walk (walk seconds, medians): original 1.280, previous 1.164, candidate 1.198.

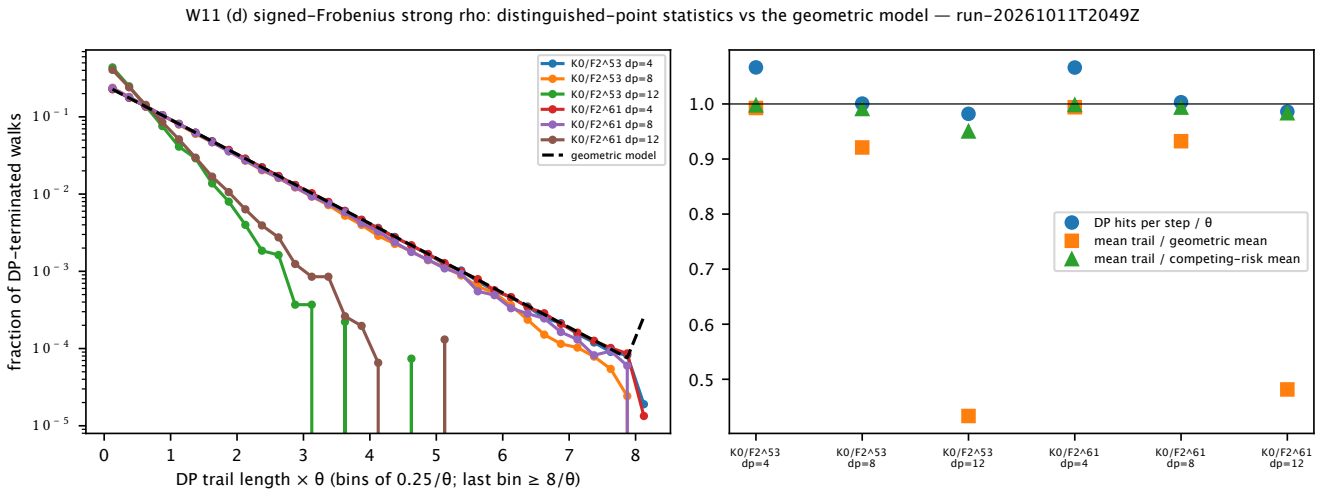
1.3. (c) k = 2 pairing loops

`raw/biext_k2.jsonl`: values of the three loops identical at every size and equal to `eds_mov::modified_tate` at (1283, 107); raw cubical value = square of the pairing on every instance.



1.4. (d) DP statistics

`raw/dp_stats.jsonl` (K0/F2^53: 64 solves per dp; K0/F2^61: 32 per dp).



1.5. (e) Challenge ladder

`raw/ladder_derive.jsonl` (levels 30–60), `raw/ladder_solve.jsonl` (720 verified solves). Pooled iterations/√(πn/4): mean 1.077, s.d. 0.546, s.e. 0.020, median 0.998.

